Practical - 14

Practical Title: Admin Dashboard with Role-Based Access Control, Dynamic Menu, and Audit Log

Problem Definition: Develop an admin dashboard that displays user statistics, student count, event count and latest registrations. Restrict access to admin users only and show dynamic menu options based on role.

A. Theory

1. Are dashboard values loaded dynamically from the database?
Answer: Yes, counts for students, events, and users are calculated via dynamic COUNT(*) SQL queries.

2. Is access restricted to admin users?
Answer: Yes, administrative panels and sensitive views are protected behind role checks.

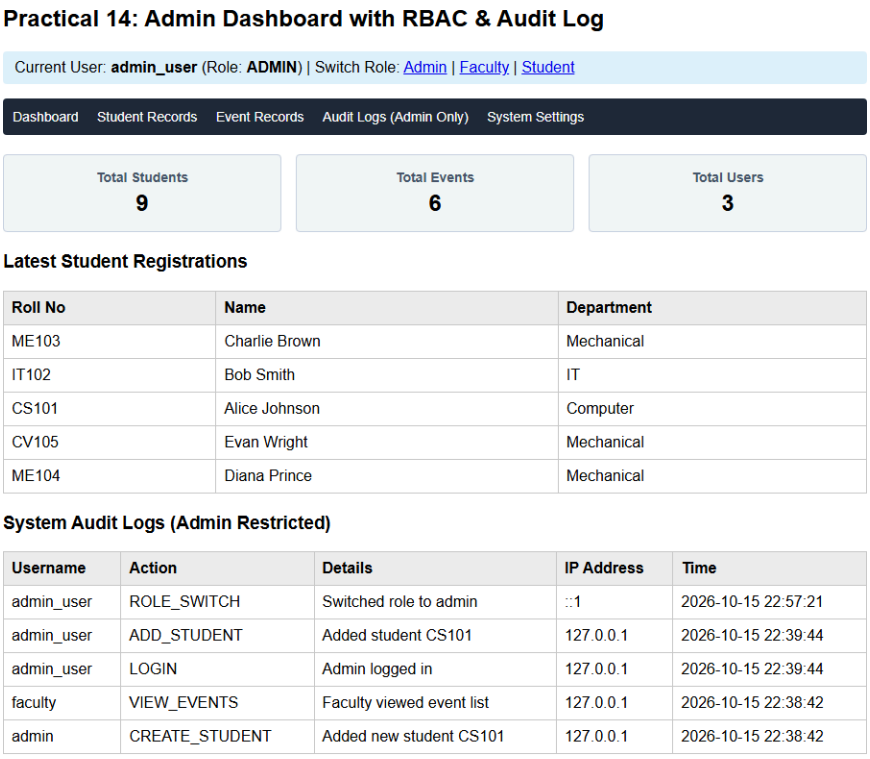
3. Are role-based menus working?
Answer: Yes, navigation elements render conditionally based on the active role (admin, faculty, student).

4. Are important admin actions recorded in audit_logs?
Answer: Yes, user actions, IP addresses, and event details are written into the audit_logs table.

B. Implementation/ Output/Result

Feature	Implementation
KPI metrics	Counts users, students, and events using COUNT(*).
Recent data	Displays latest 5 registrations using LIMIT 5.
Access control	Restricts sensitive views to the admin role.
Dynamic Menu	Renders links conditionally by session role.
Audit log	Inserts actions and IP addresses into audit_logs.

C. Evidence of Work (Screenshots)



Practical 14: Admin Dashboard with RBAC & Audit Log

Current User: **admin_user** (Role: **ADMIN**) | Switch Role: Admin | Faculty | Student

Dashboard | Student Records | Event Records | Audit Logs (Admin Only) | System Settings

Total Students	Total Events	Total Users
9	6	3

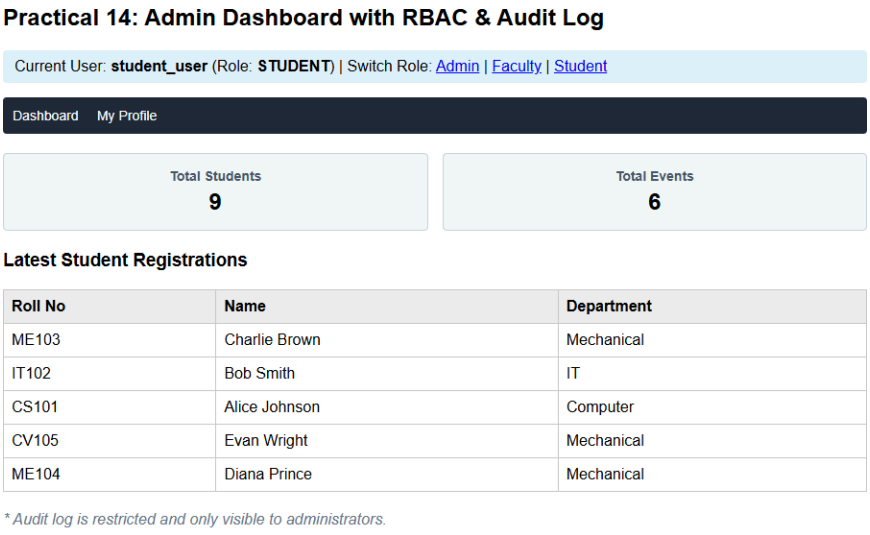
Latest Student Registrations

Roll No	Name	Department
ME103	Charlie Brown	Mechanical
IT102	Bob Smith	IT
CS101	Alice Johnson	Computer
CV105	Evan Wright	Mechanical
ME104	Diana Prince	Mechanical

System Audit Logs (Admin Restricted)

Username	Action	Details	IP Address	Time
admin_user	ROLE_SWITCH	Switched role to admin	::1	2026-10-15 22:57:21
admin_user	ADD_STUDENT	Added student CS101	127.0.0.1	2026-10-15 22:39:44
admin_user	LOGIN	Admin logged in	127.0.0.1	2026-10-15 22:39:44
faculty	VIEW_EVENTS	Faculty viewed event list	127.0.0.1	2026-10-15 22:38:42
admin	CREATE_STUDENT	Added new student CS101	127.0.0.1	2026-10-15 22:38:42

Description: Figure 14.1 admin dashboard with audit log



Practical 14: Admin Dashboard with RBAC & Audit Log

Current User: **student_user** (Role: **STUDENT**) | Switch Role: Admin | Faculty | Student

Dashboard | My Profile

Total Students	Total Events
9	6

Latest Student Registrations

Roll No	Name	Department
ME103	Charlie Brown	Mechanical
IT102	Bob Smith	IT
CS101	Alice Johnson	Computer
CV105	Evan Wright	Mechanical
ME104	Diana Prince	Mechanical

** Audit log is restricted and only visible to administrators.*

Description: Figure 14.2 student role (restricted)

D. GitHub Repository Link:

Link: https://github.com/prep04-yl/StudentHub-Portal

E. Challenges Faced:

Designing fine-grained privilege matrices without leaving endpoints exposed to URL parameter manipulation.

F. Learning Outcome:

I learned how to build a role-based admin dashboard with basic monitoring and traceability.

Faculty Remarks: **(Keep this blank for faculty)**